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ScientificProjectsController – Overview

Namespace	FougeraClub.Areas.Admin.Controllers
Purpose	CRUD operations, OTP workflow, approvals, printing, reporting, and notifications for Scientific Projects
Routing	Admin/[controller]/[action]

Controller Attributes

Attribute	Description
AdminAuthorize	Restricts access to admin users only
Area("Admin")	Defines controller under Admin area
Route	Defines fixed routing pattern

Controller Dependencies

Dependency	Purpose
IUnitOfWork	Database access and repositories
IWebHostEnvironment	File storage and environment access
IScientificProjectsService	OTP and signature business logic
UserManager	User and role management
IHubContext	SignalR real-time notifications
INotificationService	Role-based notification sending
IHttpContextAccessor	Session and role validation

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OTP & Signature Workflow

Step	Behavior
1	Trainer receives OTP and signs first
2	Edit disabled for Trainer after signing & Delete disabled for any User
3	Notification sent to Activity Monitor
4	Activity Monitor validates OTP and signs
5	Notification sent to Manager
6	Manager validates OTP and signs
7	Edit & delete disabled for all users
8	Project Card and Print enabled

Controller Actions

Action	HTTP	Description
Index	GET	List projects with search, date filter, pagination, AJAX support
Create	GET	Prepare new project with serial code and departments
Create	POST	Save project with file uploads and validation
Details	GET	View project details and signatures
Edit	GET	Load project for editing
Edit	POST	Update project and files
Delete	POST	Delete project if no signatures exist
Print	GET	Printable filtered list
ProjectCard	GET	Display project card
SendOtp	POST	Send OTP to selected role
ValidateOtp	POST	Validate OTP and store signature

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ScientificProjectsService – Overview

Namespace	Application.Services.Admin
Responsibility	OTP generation, validation, and signature persistence

Service Dependencies

Dependency	Purpose
IUnitOfWork	Database updates for signatures
IHttpContextAccessor	Session-based OTP tracking
ISMSForSendingOTPService	OTP delivery

AutoMapper – ScientificProjectsProfile

Source	Destination	Notes
ScientificProjectsVM	ScientificProjects	Maps form data to entity
ScientificProjects	ScientificProjectsVM	Reverse mapping enabled

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ScientificProjects Entity

Property	Type	Attributes	Description
Id	int	Key	Primary key
SerialCode	string	MaxLength(50)	Project serial number
DepartmentId	int?	ForeignKey	Related department
ProjectNameAr	string	MaxLength(200)	Arabic name
ProjectNameEn	string	MaxLength(200)	English name
ProjectIdeaAr	string?	MaxLength(200)	Arabic Description of Project Idea
ProjectIdeaEn	string?	MaxLength(200)	English Description of Project Idea
IdeaOwnerAr	string?	MaxLength(200)	Arabic Idea Owner Name
IdeaOwnerEn	string?	MaxLength(200)	English Idea Owner Name
Supervisor1-Supervisor6	string?	MaxLength(200)	Supervisor Name
Participant1-Participant4	string?	MaxLength(200)	Participant Name
ProjectElement1-ProjectElement8	string?	MaxLength(200)	Project Element
ExpectedCost	double?	-	Estimated cost
InstallationRecommendations	string?	MaxLength(500)	Installation Recommendations Data
DeliveryData	string?	MaxLength(500)	Delivery cost
FilePath1-FilePath6	string?	MaxLength(200)	Images pathes

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Property	Type	Validation	Description
ProjectNameAr	string	Required, Regex	Arabic project name
ProjectNameEn	string	Required, Regex	English project name
DepartmentId	int?	Required	Department selection
ProjectIdeaAr	string?	MaxLength(200)	Arabic Description of Project Idea
ProjectIdeaEn	string?	MaxLength(200)	English Description of Project Idea
IdeaOwnerAr	string?	MaxLength(200)	Arabic Idea Owner Name
IdeaOwnerEn	string?	MaxLength(200)	English Idea Owner Name
Supervisor1-Supervisor6	string?	MaxLength(200)	Supervisor Name
Participant1-Participant4	string?	MaxLength(200)	Participant Name
ProjectElement1-ProjectElement8	string?	MaxLength(200)	Project Element
ExpectedCost	double?	-	Estimated cost
InstallationRecommendations	string?	MaxLength(500)	Installation Recommendations Data
DeliveryData	string?	MaxLength(500)	Delivery cost
File1-File6	IFormFile	Image validation	Attachments
FilePath1-FilePath6	string?	MaxLength(200)	Images pathes

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Views Summary

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View	Purpose
Index	Listing, filtering, printing, Excel export
Create	Project creation
Edit	Project modification
Details	Read-only view and OTP signing
ProjectCard	Final approved project card